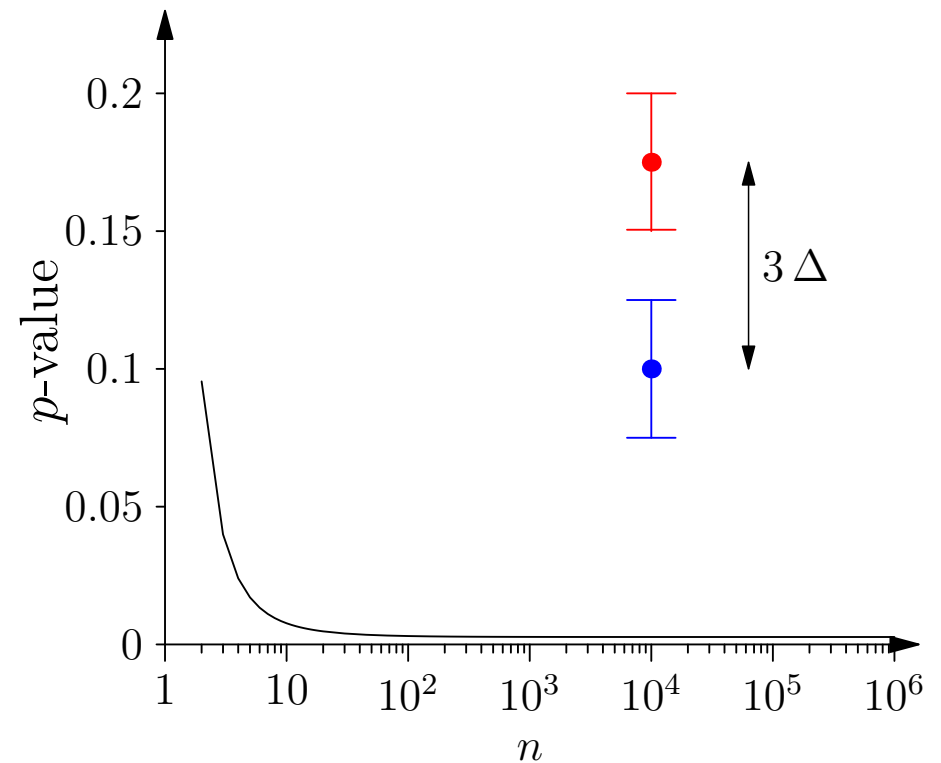
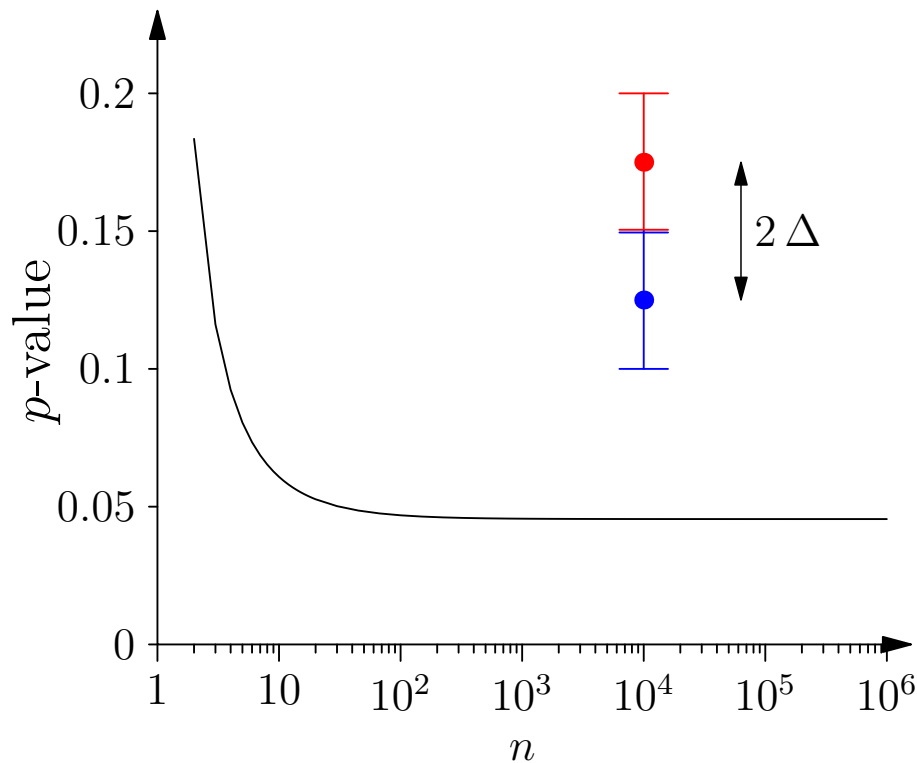


Mathematics of Machine Learning

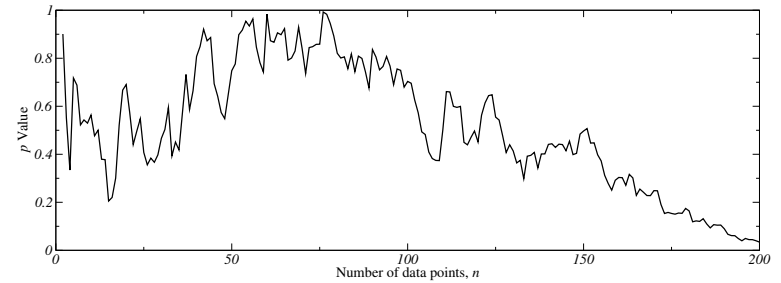
On Becoming a Scientist



interpreting data, standard errors, confidence tests

Outline

1. **Outline**
2. Experimental Error
3. Confidence Estimates



Experiments and Reproducibility

- One of the fundamental requirements of science is that experiments can be reproduced■
- This applies to machine learning and deep-learning as it does to particle physics, molecular biology, etc.■
- We need to describe completely what procedures were carried out■
- However, experiments have natural variability so what do we mean by repeatability?■

Describing Research

- When describing research you need to find a story (narrative) — why is the work you are doing interesting
- You need to give a general motivation as well as a more detailed explanation of what you are specifically going to test
- You then need to give enough details that someone can take your report/paper and reproduce what you are doing
- These days it is also common practice to put the model you are using on-line — but you should still describe everything in your report

Describing Your Experiments

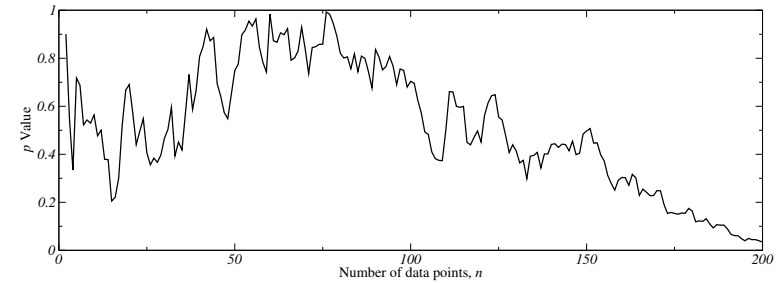
- You need to tell the reader
 - ★ What your deep-learning architecture looks like■
 - ★ Your training set■
 - ★ How you partitioned your training set into train, validation and test sets■
 - ★ Any preprocessing you performed■
 - ★ The optimiser you used, mini-batch size, learning rate■
 - ★ The number of epochs you trained for■
 - ★ Any hyper-parameter tuning you carried out■
 - ★ How long it took in clock time on the platform you are using■
- You can put some of this into an appendix if necessary■

Results

- A paper builds a story (thesis) based on experimental evidences■
- It is not enough to present results, it is also your job to interpret the results■
- Your results provide evidence in support of your thesis■
- But results all come with uncertainty. If you tested on a different test set, or used a different training set then the answer would be different■
- **Results are meaningless unless the uncertainty is quantified!■**

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Experimental Error

- Any empirical result is useless without some estimate of error■
- If you tell me algorithm A has a performance of 95% and algorithm B has a performance of 97%, how do I know that this didn't happen by chance?■
- Maybe algorithm A is actually better, but you got lucky (or unlucky)■
- We need to get a handle on the estimated experimental error■

Datasets

- We assume that our test set consists of typical examples that we might encounter and that they are independent (e.g. we are not seeing the same example over and over again)■
- These are assumptions that are not always true■
- We need to think about it and if they are not true you may need to collect new data or be very cautious of the claims that you are making■
- However, if the person curating the training set did a good job then usually these are reasonable assumptions■

Collecting Statistics

- Given some experimental measurements $\{X_i | i = 1, 2, \dots, n\}$ we can estimate the mean and variance

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n X_i \qquad \hat{\sigma}_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \hat{\mu}_n)^2$$

- These are not the true mean, $\mu = \mathbb{E}[X_i]$, and true variance $\sigma^2 = \mathbb{E}[(X_i - \mu)^2]$, but they are unbiased estimates in that

$$\mathbb{E}[\hat{\mu}_n] = \mu \qquad \mathbb{E}[\hat{\sigma}_n^2] = \sigma^2$$

Variance in Mean Estimator

- Consider the mean squared error (MSE) of the estimator $\hat{\mu}_n$

$$\begin{aligned}\mathbb{E}[(\hat{\mu}_n - \mu)^2] &= \mathbb{E}\left[\left(\left(\frac{1}{n}\sum_{i=1}^n X_i\right) - \mu\right)^2\right] = \mathbb{E}\left[\left(\frac{1}{n}\sum_{i=1}^n (X_i - \mu)\right)^2\right] \\&= \frac{1}{n^2}\mathbb{E}\left[\left(\sum_{i=1}^n (X_i - \mu)\right)\left(\sum_{j=1}^n (X_j - \mu)\right)\right] \\&= \frac{1}{n^2}\mathbb{E}\left[\sum_{i=1}^n (X_i - \mu)^2 + \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n (X_i - \mu)(X_j - \mu)\right] \\&= \frac{1}{n^2}\sum_{i=1}^n \left(\mathbb{E}[(X_i - \mu)^2] + \sum_{\substack{j=1 \\ j \neq i}}^n \mathbb{E}[X_i - \mu]\mathbb{E}[X_j - \mu]\right) \\&= \frac{1}{n^2}\sum_{i=1}^n \sigma^2 = \frac{1}{n}\sigma^2\end{aligned}$$

Estimated Error in the Mean

- The root mean squared error (RMSE) is equal to

$$\sqrt{\mathbb{E}[(\hat{\mu}_n - \mu)^2]} = \frac{\sigma}{\sqrt{n}}$$

- The **estimated error in the mean** or **standard error** is defined as $\Delta = \frac{\hat{\sigma}_n}{\sqrt{n}}$ where $\hat{\sigma}_n$ is the empirical estimate of the standard deviation
- Traditionally we would write an experimental result as $\hat{\mu}_n \pm \Delta$
- Also we only quote $\hat{\mu}_n$ up to the number of significant figures consistent with Δ
- Not 1.6582356 ± 0.122543 but 1.7 ± 0.1

Repeated Bernoulli Trials

- Suppose we are measuring a binary outcome, success or failure■
- Assuming the trials are independent (so called Bernoulli trials) then the only unknown is the probability, p , of a success■
- If we make n trials then probability of k successes is given by the binomial distribution

$$\mathbb{P}(K = k) = \binom{n}{k} p^k (1 - p)^{n-k} \blacksquare$$

- The mean and variance of K are

$$\mathbb{E}[K] = np \qquad \mathbb{E}[(K - np)^2] = np(1 - p) \blacksquare$$

Mean Squared Error

- For the case of n Bernoulli trials we can estimate the mean success rate

$$\hat{p} = \frac{K}{n}$$

- The mean squared error (MSE) in this estimate is

$$\begin{aligned}\mathbb{E}[(\hat{p} - p)^2] &= \mathbb{E}\left[\left(\frac{K}{n} - p\right)^2\right] = \frac{1}{n^2}\mathbb{E}[(K - np)^2] \\ &= \frac{1}{n^2}np(1 - p) = \frac{p(1 - p)}{n}\end{aligned}$$

Standard Error for Bernoulli Trials

- The estimated error in the mean is given by the root mean squared error (RMSE) $\sqrt{\mathbb{E}[(\hat{p} - p)^2]} = \sqrt{\frac{p(1-p)}{n}}$
- We don't know p , but we can use an estimate, $\hat{p} = \frac{K}{n}$

$$\Delta \approx \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

- We can also use a worst-case bound (when $p = \frac{1}{2}$)

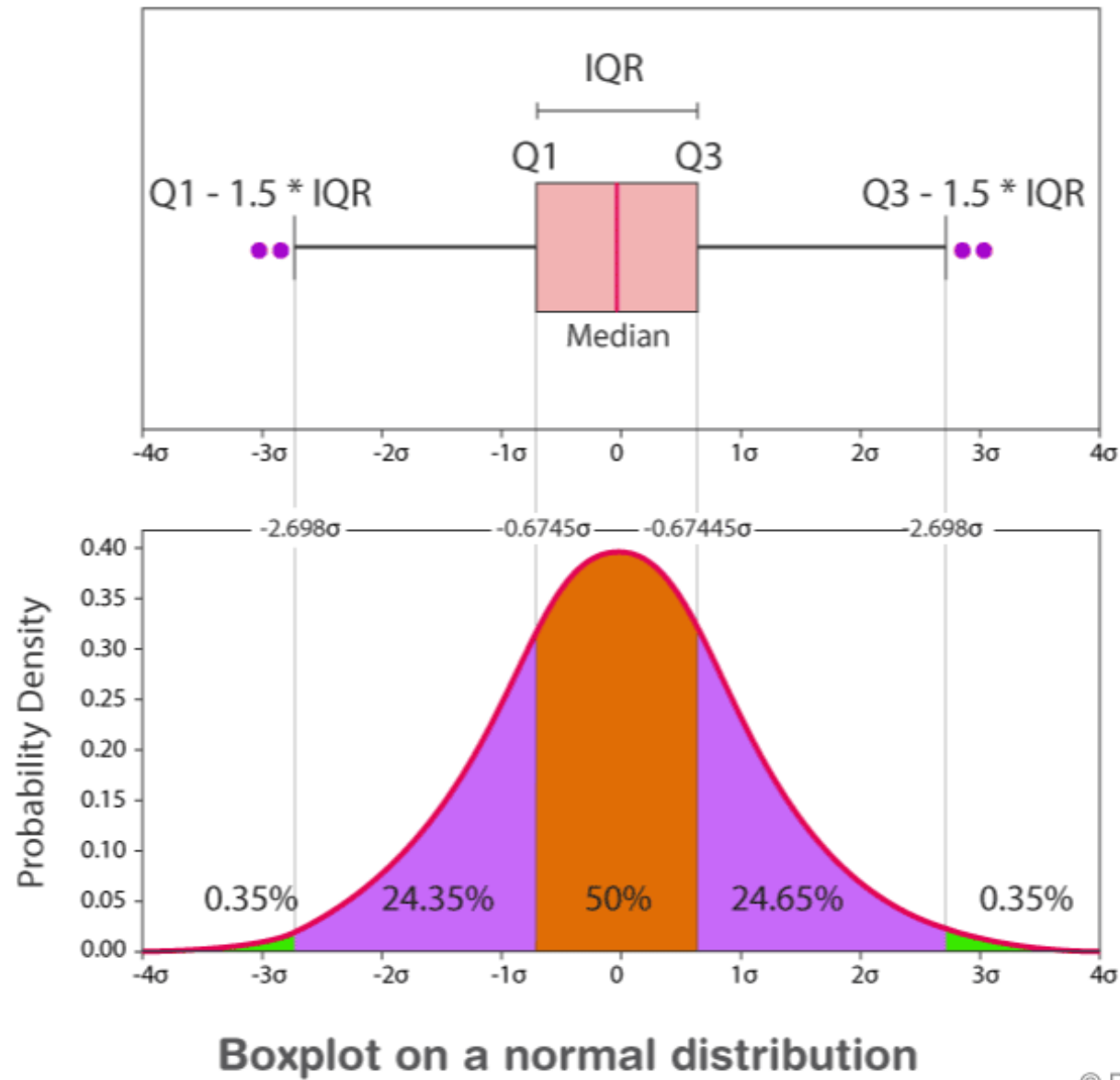
$$\Delta \approx \frac{1}{2\sqrt{n}}$$

This is pessimistic if p is small, although the previous estimate is optimistic if $\hat{p} = 0$

Errorbars

- There are different conventions on errorbars■
- Traditionally errorbars show the estimated error in the mean (standard error, $\Delta = \hat{\sigma}_n / \sqrt{n}$)■
- This means that about 30% of the time the true mean would be outside the errorbar■
- Some people use 95% confident intervals as errorbars, saying 95% of the time the true mean lies within the errorbars■—this corresponds to errorbars showing twice the standard error■
- Some people show the standard deviation as errorbars■, this tells you nothing about the confidence of the point you plot■—in my opinion it is just wrong■

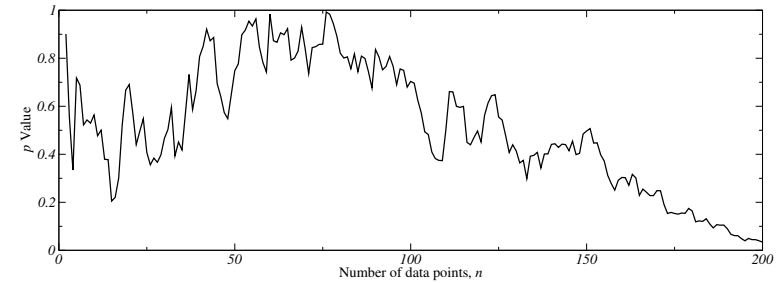
Box Plots



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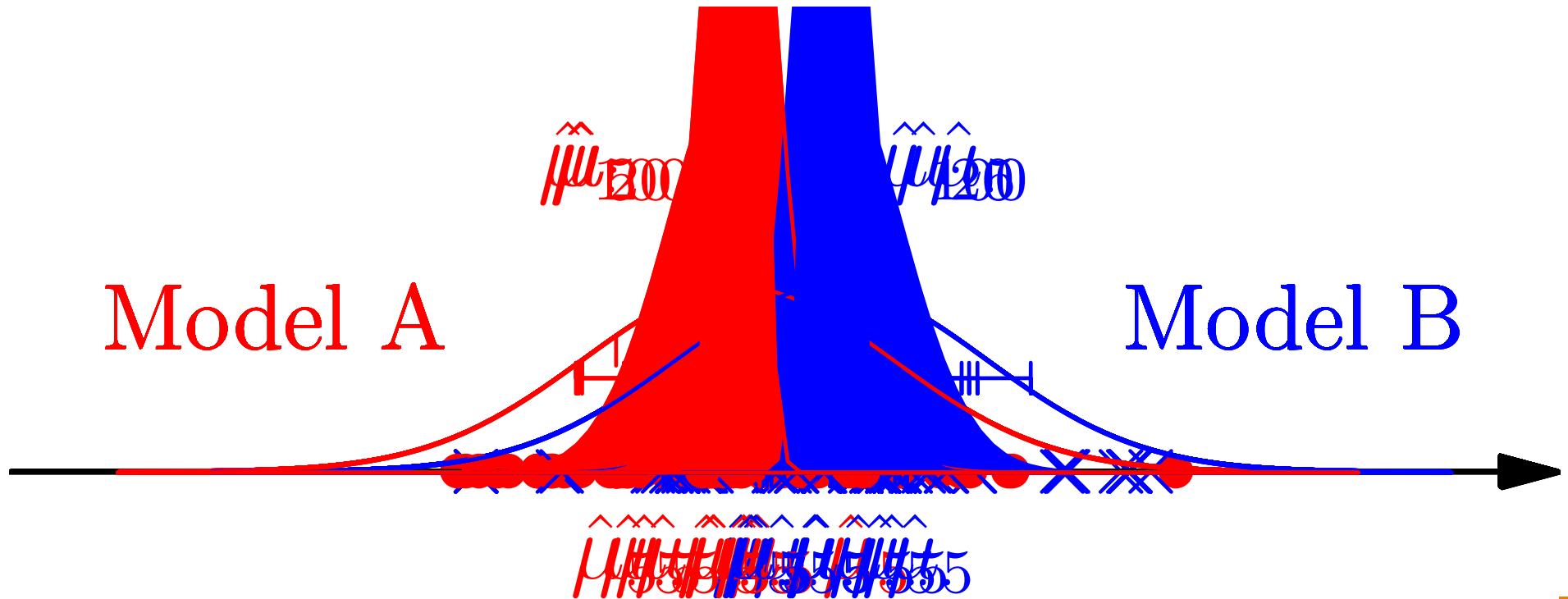
Outline

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Know What's Best

- How do we know if model B is better than model A?



The more data we have the less likely we will make a mistake

Quantifying Uncertainty

- We want a way of quantifying the chance that model B is better than model A■
- To do this we can consider a **null hypothesis**

We suppose that model A and B have the same performance■

- We then ask, under the null hypothesis, what is the chance of obtaining a gap in performance at least as big as the one we observed■
- This will depend on the number of observations and the empirical variance■

T-tests

- The original t-test was invented by William Gosset who worked for the Guinness brewery in Dublin, but published under the pseudonym *Student*■
- He worked with the null hypothesis that the distributions for A and B are normal with the same means and standard deviations■
- The slightly more common test is Welch's t-test where the null hypothesis is that the two datasets are normally distributed with the same means but different standard deviations■
- The gap is equal to the t value

$$t = \frac{\hat{\mu}_A - \hat{\mu}_B}{\hat{\sigma}_{A,B}} \blacksquare$$

Welch's T-Test

- For Welch's t-test

$$t = \frac{\hat{\mu}_A - \hat{\mu}_B}{\hat{\sigma}_{A,B}} \quad \hat{\sigma}_{A,B} = \sqrt{\frac{\hat{\sigma}_A^2}{n} + \frac{\hat{\sigma}_B^2}{m}}$$

- Where p depends on t and the *number of degrees of freedom*

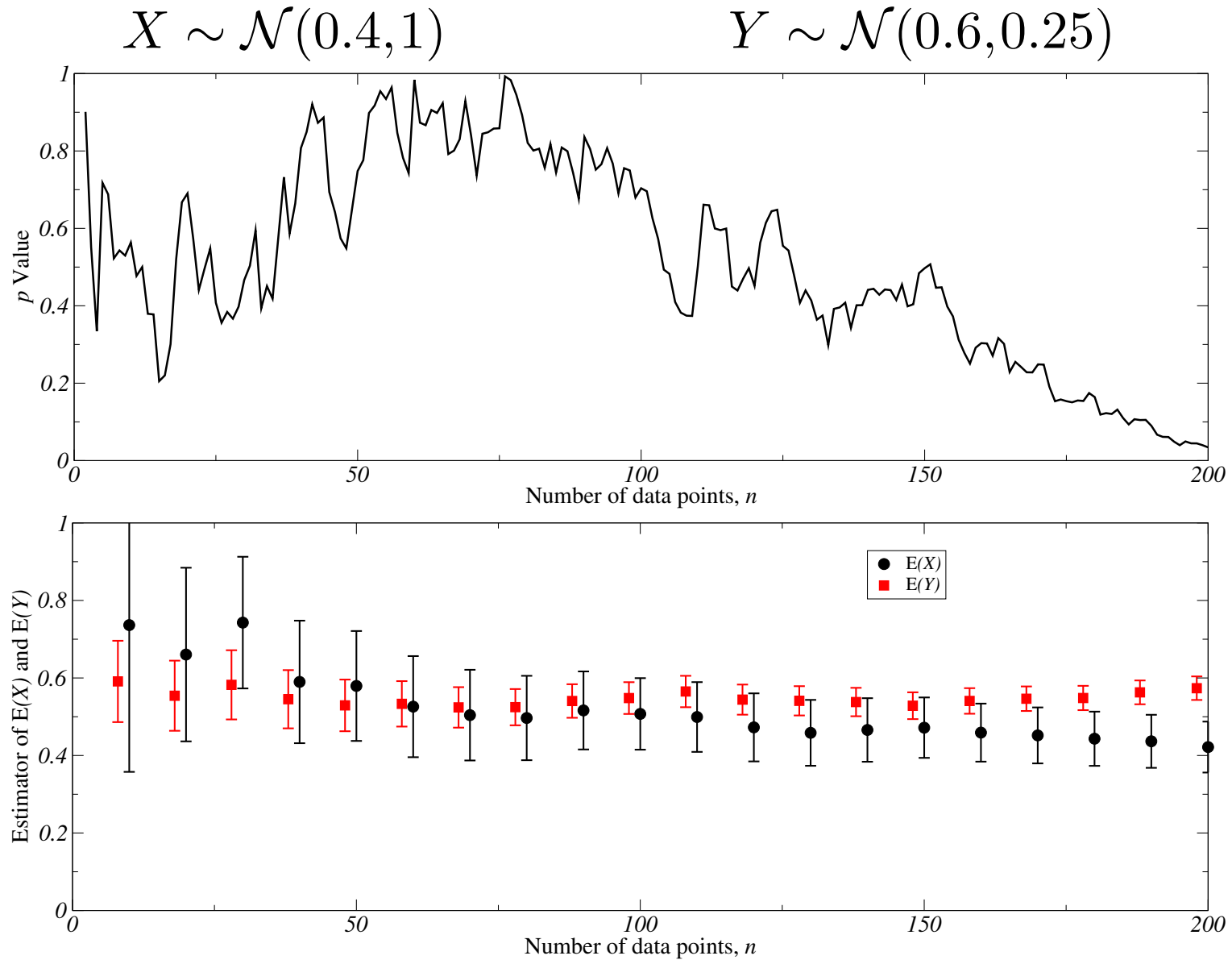
$$\nu = \frac{\hat{\sigma}_{A,B}^2}{(\hat{\sigma}_A^2/n)^2/(n-1) + (\hat{\sigma}_B^2/m)^2/(m-1)}$$

- The beauty of the t -test is that from t and ν we can compute p
- Small p is good. It shows the null hypothesis is unlikely

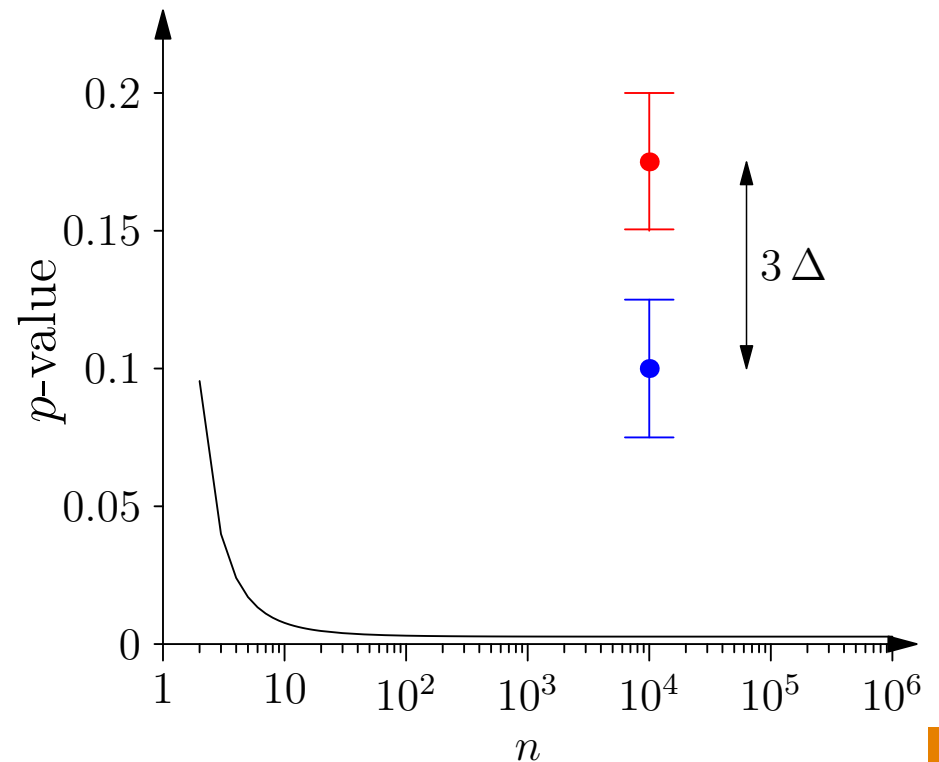
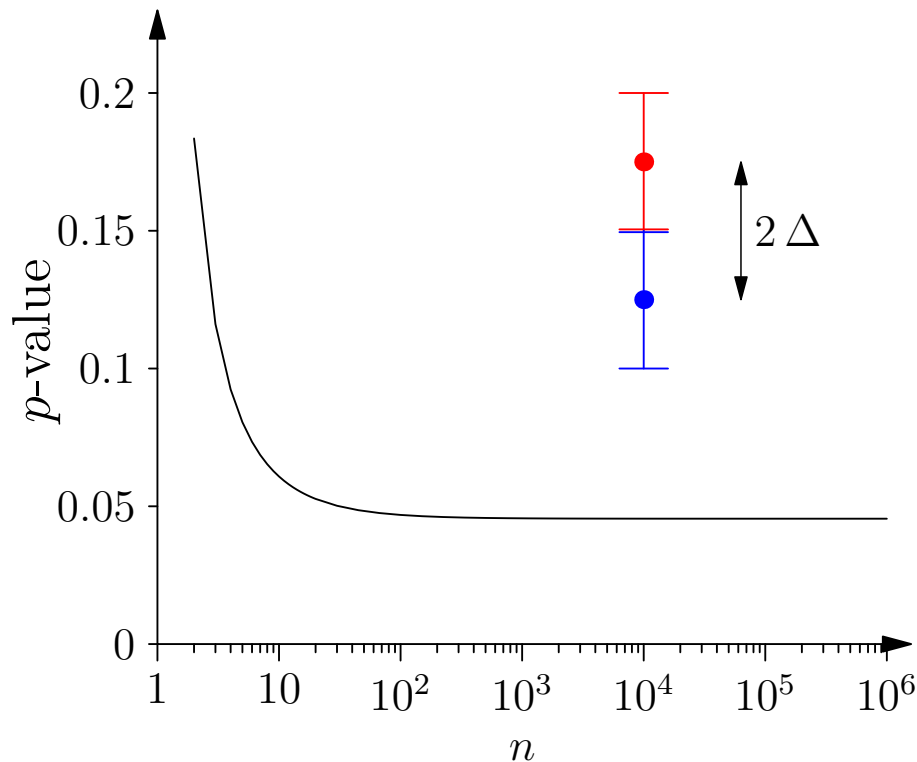
Statistical Significance

- It is common to say that A is *statistically significantly* better than B if the probability of the observed t value occurring given the null hypothesis is less than 0.05■
- Statistically significant difference in performance does not mean that the difference is anything to care about■ (that is a judgement call)■
- We are assuming the data is normally distributed■ We can check, but no one does■ Usually its not a bad approximation if the data is not normal, but there are exceptions. . . ■

Example



Error Bars and Significance



- If you have enough data points then when the errorbars (1 standard error) are separate then the two distributions are significantly different

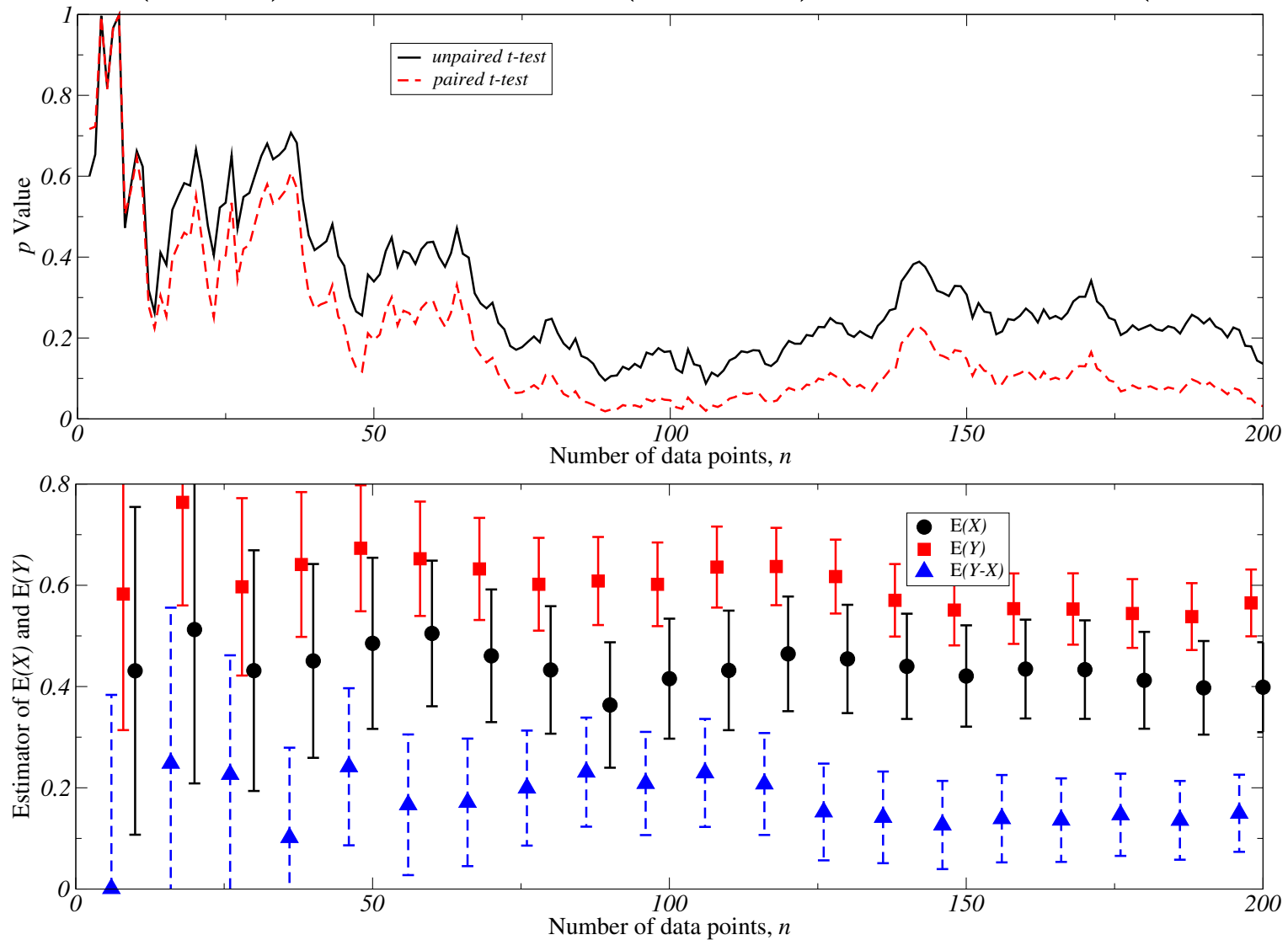
Paired T-Test

- Sometimes the fluctuations in performance will come both from the test example and from the difference in performance■
- For example model A might always be better than model B, but the performance fluctuates depending on the example■
- A normal t-test might not show any significant difference■
- Using a paired t-test we instead use

$$\hat{\sigma}_{X+Y}^2 = \frac{\hat{\sigma}_X^2 + \hat{\sigma}_Y^2 - \hat{c}_{X,Y}}{n}$$
$$\hat{c}_{X,Y} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \hat{\mu}_X)(Y_i - \hat{\mu}_Y)■$$

Paired T-Test Example

$$\begin{aligned} X &\sim Z + X_1 & Y &\sim Z + Y_1 \\ Z &\sim \mathcal{N}(0, 0.8) & X_1 &\sim \mathcal{N}(0.4, 0.2) & Y_1 &\sim \mathcal{N}(0.6, 0.2) \end{aligned}$$



Cheating Significant Tests

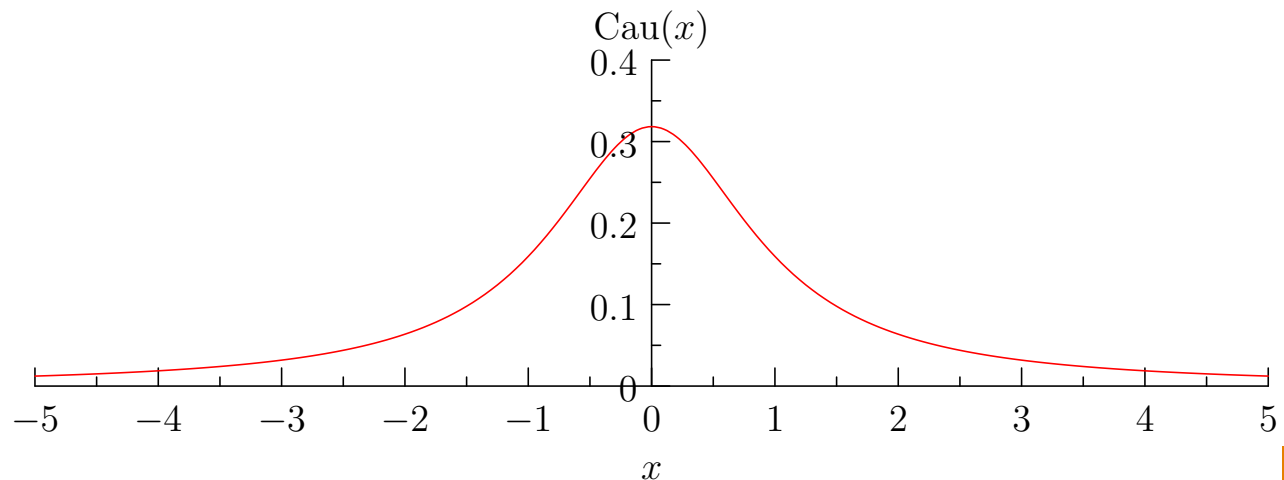
- You should only run your t-test once■
- If you run multiple t-tests until you get one with $p < 0.05$ then you have cheated■
- With a p value of 0.05 you expect to find the observed t value given the null hypothesis once every 20 trials■—if you do the t-test twice you should expect this to happen every 10 trials (i.e. $p = 0.1$)■
- If you run t -tests until you find a p value of 0.05 you are cheating■
- If you want to compare many different models then you should use ANOVA instead■

Computing Significance Ab-Initio

- The t -test results follow from a beautiful, but difficult calculation■
- However, you can simulate the null-hypothesis multiple times and measure empirically the number of times the t -value exceeds the observed t -values■
- This isn't necessary under the assumptions that the data is normally distributed (done by Student and Welch)■
- However, there are other times where you want to show that what you have found has not happened by chance■ Often it is easy to simulate a null hypothesis■

Good, Bad and the Ugly

- What I've told you is true of data that follows well behaved distributions
- Just occasionally you come across data that has long (thick) tails



- Here taking the average of many samples might not give you reliable estimates of the mean
- This isn't that common, but when nothing seems to make sense plot a histogram of the data and check for thick tails

Lessons

- You need to understand the errors in your data■
- Computing the standard error is always necessary■
- In some cases you can quantify your certainty that A is better than B using statistical confidence tests—t-tests■ (other tests are available)■
- You need to apply common sense in interpreting data■
- **The person you cheat when you don't do this is yourself■**